

Inelastic e-p scattering

$$E_e \gg 1 \text{ GeV}$$

$$e^- + p \rightarrow e^- + p \quad \text{elastic}$$

$$e^- + p \rightarrow e^- + X \quad \text{inelastic}$$

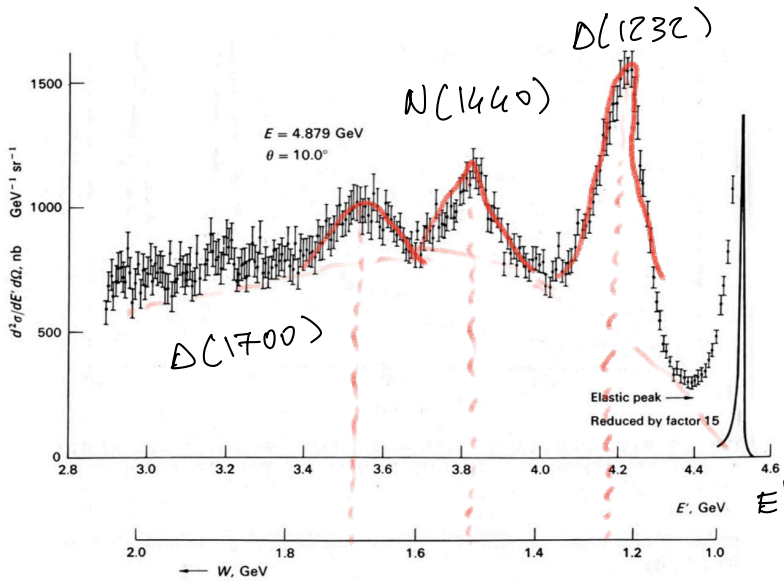
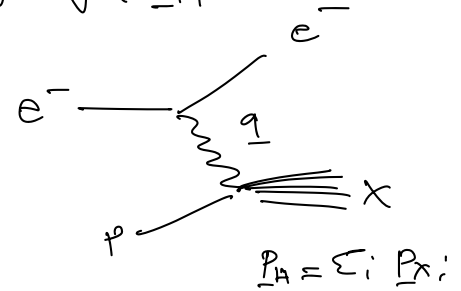


Figure 8.6 Excitation curve of inelastic ep scattering, obtained at the DESY electron accelerator (Bartel *et al.* 1968). E and E' are the energies of the incident and the scattered electron, and W is the mass of the recoiling hadronic state. The peaks due to the pion-nucleon resonances of masses 1.24, 1.51, and 1.69 GeV are clearly visible.

$$Q^2 = M^2 - w^2 + 2Mv$$

$$W = \sqrt{1 + |P_H|^2}$$



$$P_H = P_p^{\text{in}} + q$$

$$e^- + p \rightarrow e^- + X^+$$

$$e^- + (\text{Baryon})^+$$

$$e^- + \Delta(1232)$$

$$e^- + N(1440)$$

$$e^- + p \rightarrow n + \bar{u}$$

$$e^- + p \rightarrow \bar{\nu}^0$$

$$e^- + p \rightarrow \pi^+ + \pi^-$$

Δ baryons combinations.
 N baryons (u, u, d)

$$\Delta^{++} = uuu \quad \Delta^+ = uud \quad \Delta^0 = udd \quad \Delta^- = ddd$$

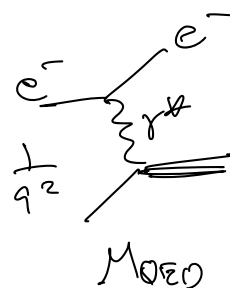
Hadronic resonances: Δ, N

$$\Gamma = \frac{1}{\tau}$$

$$B_W(M, \Gamma) \otimes G(0|\sigma)$$

lepton number
 electric charge

Baryon number



$$Q^2 \approx 1 \text{ GeV}^2$$

$$M_W^2 \approx 80 \text{ GeV}^2$$

$$\sigma_{\text{DIS}} \approx (6400) \times \sigma_{\text{weak}}$$

No worries about weak interaction

$\frac{\Gamma}{M}$ large. large width
 very small lifetime

$$\sigma \approx \sigma$$

$\Delta(1232)$ DECAY MODES

The following branching fractions are our estimates, not fits or averages.

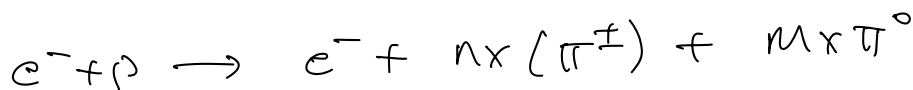
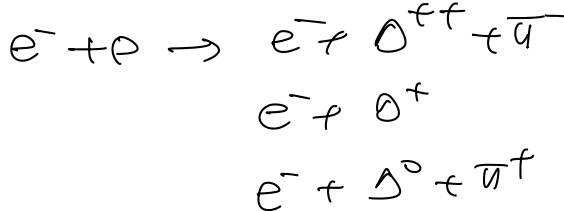
Mode	Fraction (Γ_i/Γ)
Γ_1 $N\pi$	99.4 %
Γ_2 $N\gamma$	0.55-0.65 %
Γ_3 $N\gamma$, helicity=1/2	0.11-0.13 %
Γ_4 $N\gamma$, helicity=3/2	0.44-0.52 %
Γ_5 $\rho e^+ e^-$	$(4.2 \pm 0.7) \times 10^{-5}$

$N(1440)$ DECAY MODES

The following branching fractions are our estimates, not fits or averages.

Mode	Fraction (Γ_i/Γ)
Γ_1 $N\pi$	55-75 %
Γ_2 $N\eta$	<1 %
Γ_3 $N\pi\pi$	17-50 %
Γ_4 $\Delta(1232)\pi$, P-wave	6-27 %
Γ_5 $N\sigma$	11-23 %
Γ_6 $\rho\gamma$, helicity=1/2	0.035-0.048 %
Γ_7 $n\gamma$, helicity=1/2	0.02-0.04 %

$$p = (0.3) BCT RW$$



at high energy need to distinguish e^-/π^-

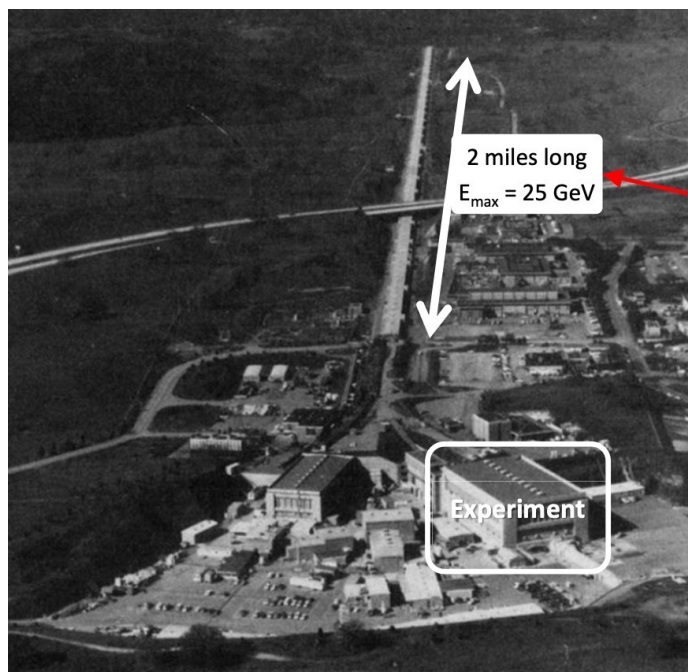
$$p = 5 \text{ GeV} \quad (\beta\gamma)_e = \frac{p}{m} = \frac{5000}{0.5} = 10^4 \quad \beta \approx 1$$

$$(\beta\gamma)_\pi = \frac{5}{0.140} = 35 \quad \beta = \frac{p}{E} < 1$$

$\Rightarrow$ Cherenkov detector $\Rightarrow$ distinguish $e^-/\pi^-/\pi^+$

$\Rightarrow$ spectrometer to measure momentum, charge

SLAC 1968 beam e^-



$E = 25$
GeV

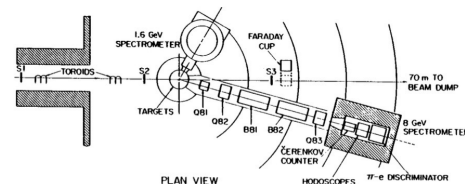
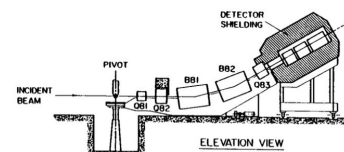
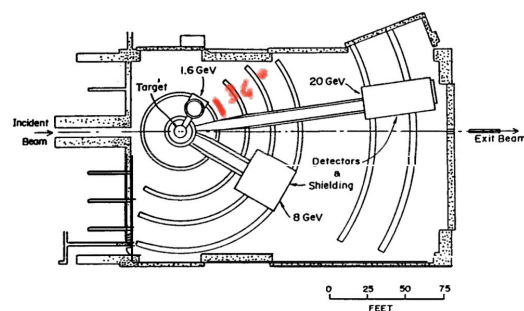


Fig. 2. (a) Plan view of End Station A and the two principal magnetic spectrometers employed for analysis of scattered electrons. (b) Configuration of the 8 GeV spectrometer, employed at scattering angles greater than 12°.

3 spectrometers.

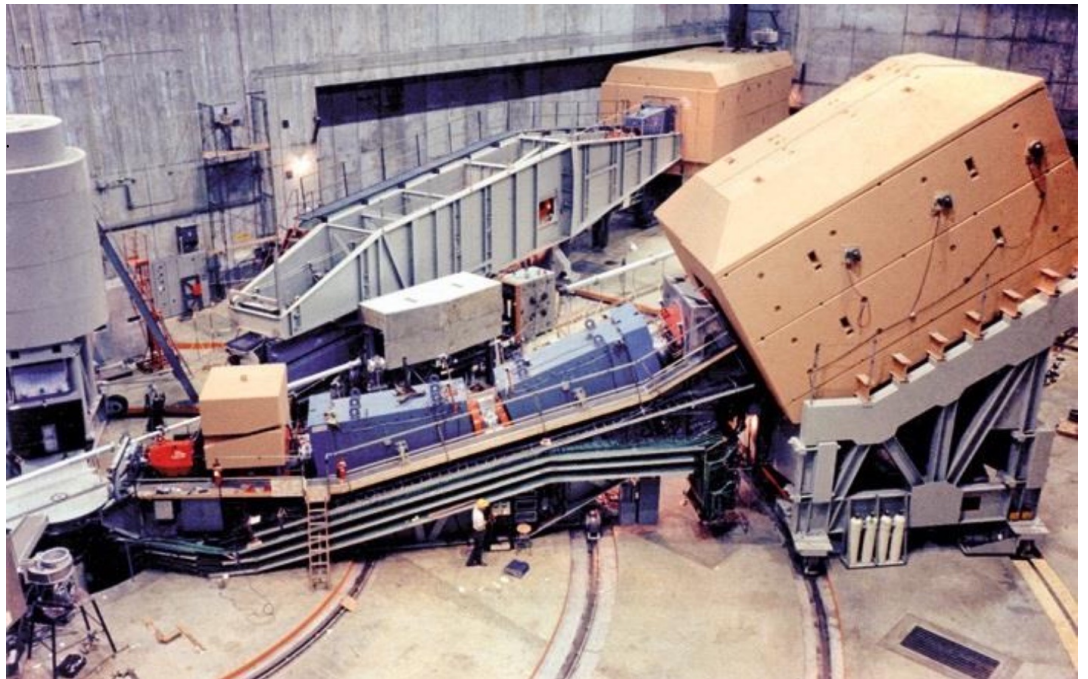
$\theta = 34^\circ$ small one

other 2

$\theta > 12^\circ$

$Z_p = 1$

$Z_T = 1$



Measure $(E', \theta) \rightarrow Q^2, \nu$

$$Q^2 = 4EE' \sin^2 \frac{\theta}{2}$$

$$\nu = E - E'$$

$$\left. \frac{d\sigma}{d\Omega} \right|_{\text{measured}} = \frac{\alpha^2}{Q^4} E'^2 \frac{E'}{E} \cos^2 \frac{\theta}{2} \left[W_2(Q^2, \nu) + 2W_1(Q^2, \nu) \tan^2 \frac{\theta}{2} \right]$$

structure functions W_1, W_2

$\left(\frac{d\sigma}{d\Omega} \right)_{\text{Mott w/ recoil}}$

Measure $\left(\frac{d\sigma}{d\Omega} \right) @ (Q, E') \Rightarrow$ how to extract W_1, W_2 ?

$(E', \theta^1) \rightarrow (Q^2, \nu)$ same (Q^2, ν) from 2 different measurements.
 $(E', \theta^2) \rightarrow$

$\Rightarrow$ extract both W_1, W_2

$$N = \underbrace{\left(\frac{d\sigma}{d\Omega} \right)}_{\sigma_{\text{fiducial}}} \times (\Delta\Omega)_{\text{detector}} \times \underbrace{L \times \Delta t}_{\substack{L_{\text{int}} \\ \text{integrated luminosity}}} \quad \text{instantaneous luminosity}$$

$e^- + p \rightarrow e^- + p$ elastic

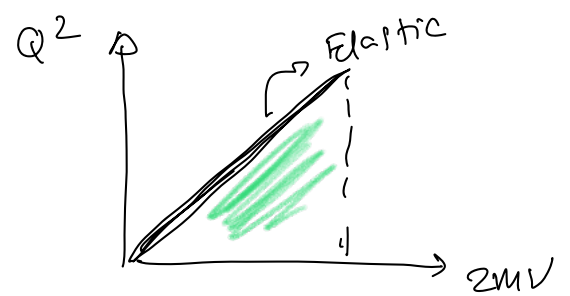
well known $\left. \frac{d\sigma}{d\Omega} \right|_{\text{Mott}}$

$N_{\text{elastic}} = L \cdot \sigma_{\text{elastic}}$
 Selection

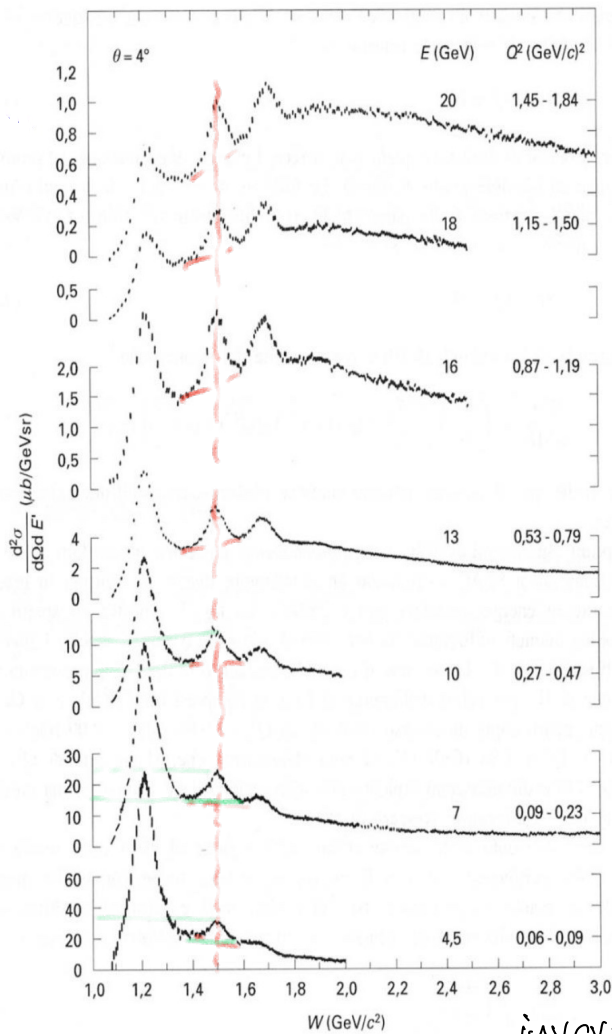
Elastic $Q^2 = 2m\nu$

L measured from elastic scattering.

$$N_{\text{inelastic events}} = \textcircled{L} \sigma_{\text{inelastic}}$$



$$\left. \frac{d\sigma}{d\Omega} \right|_{\text{inelastic measured}} = \frac{N_{\text{inelastic}}}{L} = \frac{N_{\text{inelastic}}}{N_{\text{elastic}}} \sigma_{\text{elastic}}$$



$$\frac{d^2\sigma}{d\Omega dE'} \quad \frac{\mu b}{GeV} \quad \frac{1}{sr}$$

events

(solid circle) (E' bin width)

events as function of E'

$$\theta = 4^\circ \quad Q^2 = 4E E' \sin^2 \frac{\theta}{2}$$

1) For $E \neq$ elastic peak $\rightarrow$

$$\sigma_{\text{tot}} = \sigma_{\text{el}} + \sigma_{\text{inel}}$$

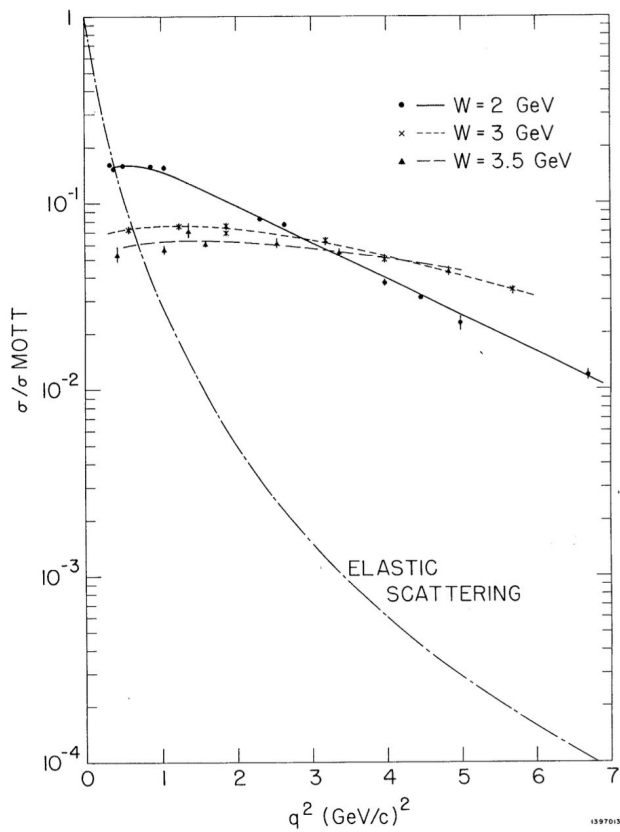
2) At fixed W $\rightarrow$ for $E \neq$

invariant mass of hadronic system.

$$\left. \frac{d\sigma}{d\Omega} \right|_{\text{inelastic}} = W_2(Q^2, \nu) + 2W_1(Q^2, \nu) \tan^2 \frac{\theta}{2} \quad @ \text{ fixed } \theta$$

$$\left. \frac{d\sigma}{d\Omega} \right|_{\text{elastic}}$$

In principle they could depend on Q^2, ν (the only kinematic variables in inelastic scattering)



$$\left. \frac{d\sigma}{d\Omega} \right|_{\text{elastic}} = \left. \frac{d\sigma}{d\Omega} \right|_{\text{Mott}} \left(1 + \frac{Q^2}{2M^2} \tan^2 \frac{\theta}{2} \right)$$

elastic scattering of e^- on Dirac proton

$$Q^2 = M^2 - m^2 + 2MV$$

$$Q^2 = 4EE' \sin^2 \frac{\theta}{2}$$

$$\frac{d\sigma}{d\Omega} \Big|_{\text{inel.}} \approx \text{constant}$$

$$\frac{d\sigma}{d\Omega} \Big|_{\text{Mott}}$$

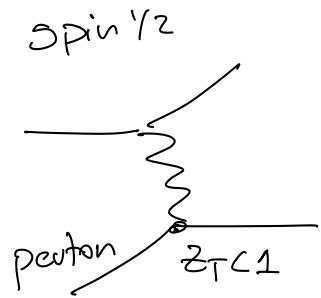
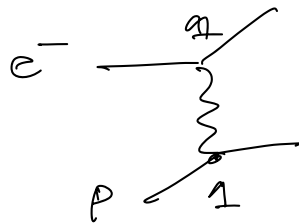
Reminder about Form factor

$$\left. \frac{d\sigma}{d\Omega} \right|_{\text{struc}} = \left. \frac{d\sigma}{d\Omega} \right|_{\text{point}} |F(Q^2)|^2$$

$$|F(Q^2)|^2 \leq 1$$

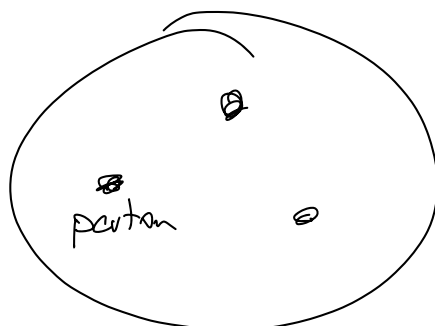
$$\frac{d\sigma}{d\Omega} = \frac{(\alpha Z_T)^2}{Q^4} (---)$$

$$Z_T < 1$$



$$\left. \frac{d\sigma}{d\Omega} \right|_{\text{proton}} = \frac{\alpha^2}{Q^4} (---) \quad \left. \frac{d\sigma}{d\Omega} \right|_{\text{neutron}} = \frac{(\alpha Z_T)^2}{Q^4} (---)$$

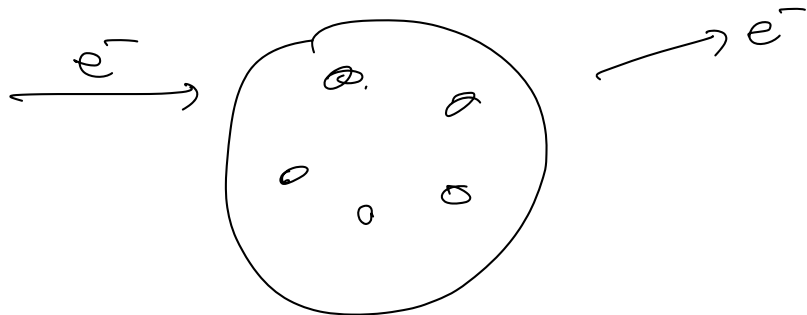
neutron: $S = 1/2$, $Z_T < 1$ fractional charge.



proton seen as container of partons.

$$e^- + p \rightarrow e^- + X \Rightarrow e^- + X \rightarrow e^- + X$$

↪ parton



$$| \text{Diagram 1} |^2 = | \text{Diagram 2} |^2 + | \text{Diagram 3} |^2 + | \text{Diagram 4} |^2$$

Elastic proton- e^- scattering $\equiv$ incoherent sum of e^- -parton scattering.

Quark-parton Model

$$e^- + p \rightarrow e^- + X$$

$$\left. \frac{d\sigma}{d\nu} \right|_{ep \text{ inelastic}} = \frac{\alpha^2}{Q^4} E'^2 \frac{E}{E'} \left[W_2(Q^2, \nu) \cos^2 \frac{\theta}{2} + 2W_1(Q^2, \nu) \sin^2 \frac{\theta}{2} \right]$$

$$\left. \frac{d\sigma}{d\nu dE'} \right|_{\text{parton}} = \frac{\alpha^2}{Q^4} E'^2 \frac{E}{E'} z_q^2 \left[\cos^2 \frac{\theta}{2} + \frac{2Q^2}{4m^2} \sin^2 \frac{\theta}{2} \right] \delta \left(\nu - \frac{Q^2}{2m} \right)$$

mass parton $m = xM$

M = mass of proton.



Elastic scattering
 e^- -parton

$$\delta \left(\nu - \frac{Q^2}{2m} \right) \Rightarrow \nu = \frac{Q^2}{2m} = \frac{Q^2}{2xM} \Rightarrow x = \left(\frac{Q^2}{2M\nu} \right)$$

x : fraction of proton mass assigned to the parton.

$$W_1(Q^2, \nu) = \frac{Q^2}{4m^2} z_q^2 \delta \left(\nu - \frac{Q^2}{2m} \right) \quad x = \frac{Q^2}{2M\nu} \left(= \frac{m}{M} \right)$$

$x \in (0, 1] \quad m \in [0, M]$

parton density function $f(x)$

$$\int_0^1 f(x) dx = 1 \quad f(x) dx = \text{prob. of having a parton with mass between } [xM, (x+dx)M]$$